DERMONECTIN®

ACTIVE PRECURSOR OF FIBRONECTIN
MOISTURIZATION FUNCTIONALIZER

Vevy Codex 18.1926

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DERMO COSMETIC GRADE

vevy europe
OUR CHEMISTRY IS YOUR CHEMISTRY

DERMONECTIN® Active precursor of fibronectin: Moisturization Functionalizer

WHAT IS IT?

Dermonectin® (Vevy codex 18.1926) is a low molecular weight precursor of fibronectin. Characteristic for its basic aminoacid sequence.

CHEMICAL STRUCTURE

Its INCI Name is Oligopeptide-5.

FIBRONECTIN

High molecular weight glycoprotein of plasma and extracellular matrix.
In dermis it binds several extracellular matrix components such as collagen, fibrin, proteoglycans and elastic fibers, thus promoting a proper skin moisturization.
Giving its several binding sites, fibronectin plays a major role in cell adhesion, growth, migration and differentiation.
It is fundamental for the wound healing process, indeed fibronectin has a crucial role promoting wound contraction and determining the formation of a proper substratum for migration and growth of cells during the development and organization of granulation tissue.

PRODUCT SPECIFICATIONS

NOMENCLATURE	
Trade Name	DERMONECTIN®
Vevy Codex	18.1926
Description	First effective oligomer precursor of fibronectin which, as such, really helps increasing skin firmness and resiliency. Vehiculated in propylene glycol or butylene glycol.
Total Quality Assurance	DCG [Dermo&Cosmetic Grade]
USE	
Use Levels	3% - 6% in O/W, W/O and W/O/W emulsions, gels, water and water/glycerol systems.
Properties	Cutaneous cohesion factor. Basic active principle for the treatment of skin elasticity. The use of DERMONECTIN® confers a significant increase in elasticity and moisture, a perceptible decrease in the number of small wrinkles and in the size of the larger ones. Available DERMONECTIN® P-F (Vevy codex 18.5085) without parabens too.
IDENTIFICATION	
Appearance	Clear liquid
Odour	Characteristic
Colour	Straw yellow to dark yellow
Solubility	Water and water/glycerol systems

A SAFE PRODUCT

TOXICOLOGICAL PROTOCOL AVAILABLE

STUDIES

SKIN ABSORPTION

Labelled ^{14}C biologically active tetrapeptide of **Dermonectin®**: L-Arginine, L-Glycine, L-Aspartic acid, L-Serine.

Experimental Data

- Application of 1g of a 5% **Dermonectin®** O/W emulsion;
- Radiometric evaluation on nude mice.

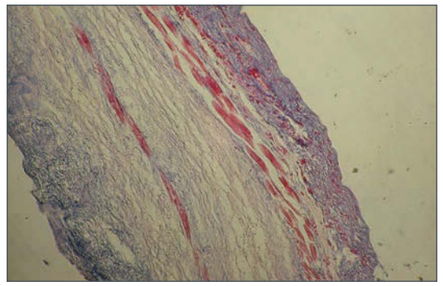
Results

Labelled **Dermonectin®** penetrates the epidermal barrier and reaches the dermis (0.2 g of cream shown in the dermis).

EFFECT ON SKIN TROPHISM

Untreated sample

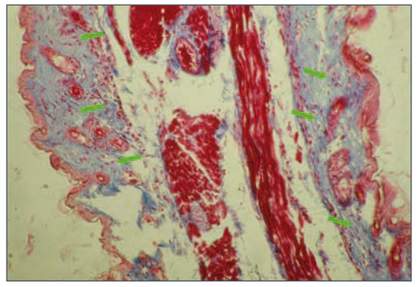
Untreated sample of newborn mice. Normal appearance of cutaneous surface.



Azan-Mallory (120x).

Treated sample with Dermonectin®

Treated sample with **Dermonectin®** at 5% for 10 days.
The epidermis covers a well developed dermis rich in collagen fibers, as shown by the specific coloration (blue).



Azan-Mallory (250x)

The fibroblasts (dark spots in the dermis) are the cells that mostly get advantage of the presence of **Dermonectin®** and are responsible for the well-being state of the dermis.

EFFECT ON WOUND HEALING

Experimental Data

- 5% **Dermonectin®** O/W emulsion;
- Topical pre-treatment of 30 days;
- Skin incision (surgery) on the 31st day.

Results

Wound healing process accelerated by more than 25% when **Dermonectin®** is applied on the skin.

EFFECT ON XYLENE-INDUCED DERMATITIS

Xylene induces desquamation of epidermis and a thickening and loss of elasticity of dermis.

Experimental Data

Induction of scleroderma-like alteration of mice skin with repeated treatment with xylene:

- 5% **Dermonectin®** O/W emulsion;
- Application twice a day;

Results

Reduction of recovery-time by 30.5% compared to controls; **Dermonectin®** has a dermotrophic effect and it is effective in the *restitutio ad integrum* process.

CLINICAL EVALUATION (DOUBLE-BLIND, DERMATOLOGIST EVALUATION)

Experimental Data

- 5.6% **Dermonectin®** O/W emulsion;
- Daily application;
- 25 volunteers (45-55 years old);
- 60 days;

Evaluated parameters: skin wrinkling, moisturization, elasticity, appearance.

Results

Increase in elasticity and moisturization;
Reduction in the number of fine lines and dimension of wrinkles.

EXAMPLES OF FORMULATIONS

ELASTICIZING EYE CONTOUR GEL	%
Demineralized Water	87.85
Hyaluramine®-S	0.50
Dermonectin®	**6.00**
Undebenzofene®-C	0.80
Poliglicoleum®	3.20
Hydroessential Flores Rosae	0.15
Liporamnosan	1.50
	Total 100.00

MATRIX SUPPORT EMULSION	%
Xalifin® 15 N-P	12.0
Trioxene®-LV	3.0
Filagrinol® N-P	7.0
Salycuminol®	1.0
Cetacene®	2.0
Undebenzofene®-C	1.3
Demineralized Water	58.0
Propilenglicole DCG	2.0
Hyaluramine®-S	0.5
Collagenon®	5.0
Dermonectin®	**3.0**
Idroramnosan	1.0
Xalidrene®	4.0
Parfum	0.2
	Total 100.0